DAY-3
DOCUMENTATION

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1.objective:

To understand python functions and Lists ,learn how to create and use functions, perform list operations, develop simple programs using functions and lists.

Functions:

A function is a reusable block of code that performs a specific task. Function make programs easier to read and understand.

Syntax:

```
def  function_name():
```

Types of Functions:

- Bulit-in functions

  Print()

  Input()

  Len()

  Sum()

- User defined functions

  Function without parameters

  Function with parameters

Function with return value

Why functions:

- Code reusability
- Reduces repetition
- Improves Readability
- Easy maintance

Advantages of function:

- Saves time
- Avoids duplicate code
- Easy debugging
- Better code management
- Easy modification

Lists:

A list is a collection used to store multiple values in a single variable.

Syntax:

list_name = [item1, item2, item3]

Features of Lists:

- Stores multiple values in one variable.
- Lists are ordered.
- Lists are mutable(values can be change).
- Lists can store different data types.
- Lists allow duplicate values.

Common List operations:

1. append() :

   Adds an item to the end of the list.

2. remove() :

Removes a specific item from the list.

3. insert() :

inserts an item at a specific position.

4. sort() :

sorts the list in ascending order.

5. len() :

returns the number of items in a list.

Why Lists :

- Store multiple values.
- Easy data management.
- Supports loops and iterations.
- Easy to add, remove and update data.
- Reduces the need for multiple variables.

Advantages of Lists:

- Easy to use.
- Dynamic in size.
- Supports different data types.
- Useful for storing large amount of data.

Programs:

Q1. Multiplication table generator using functions

Aim: To generate a multiplication table using a function.

Program:

```
def multiplication_table (num):
    for i in range(1,11):
```

```
print(num, "X", i, "=",num*i)
n =int(input("Enter a number:"))
multiplication_table(n)
```

Screenshot output:

```
def multiplication_table (num):
    for i in range(1,11):
        print(num, "X", i, "=",num*i)
n =int(input("Enter a number:"))
multiplication_table(n)
```

```
Enter a number:5
5 X 1 = 5
5 X 2 = 10
5 X 3 = 15
5 X 4 = 20
5 X 5 = 25
5 X 6 = 30
5 X 7 = 35
5 X 8 = 40
5 X 9 = 45
5 X 10 = 50
```

Logic:

1. Create a function named multiplication_table () that accepts a number as a parameter.
2. Use a for loop to iterate from 1 to 10.
3. Multiply the given number with each value of the loop.
4. Display the multiplication result in table format.
5. Take a number as input from the user.
6. Call the function and pass the input number as an argument.

Q2. Create a simple data store using lists

Aim: Create a simple data store using list.

Program:

```
data =[]
data.append("mohana")
```

```
data.append("siva")
data.append("priya")
print("stored data:")
for item in data:
    print(item)
```

Screenshot output:

```
data =[]
data.append("mohana")
data.append("siva")
data.append("priya")
print("stored data:")
for item in data:
    print(item)
```

```
stored data:
mohana
siva
priya
```

Logic:

1. Create an empty list named data.
2. Add values to the list using the append() method.
3. Print a heading to display the stored data.
4. Use a for loop to access each item in the list.
5. Display all stored items one by one.

Q3. Add student names into a list

Aim: Add student names into a list.

Program:

```
students =[]
n = int(input("Enter number of students:"))
for i in range(n):
        name=input("Enter student name:")
```

```
        students.append(name)
print("Student name:")
for name in students:
    print(name)
```

Screenshot output:

```
students =[]
n = int(input("Enter number of students:"))
for i in range(n):
  name=input("Enter student name:")
  students.append(name)
print("Student name:")
for name in students:
  print(name)
```

```
Enter number of students:3
Enter student name:priya
Enter student name:kavya
Enter student name:karthik
Student name:
priya
kavya
karthik
```

Logic:

1. Create an empty list named students.
2. Read the number of students from the user.
3. Use a for loop to enter student names.
4. Store each name in the list using append().
5. Print a heading to display student names.
6. Use another for loop to display all names stored in the list.

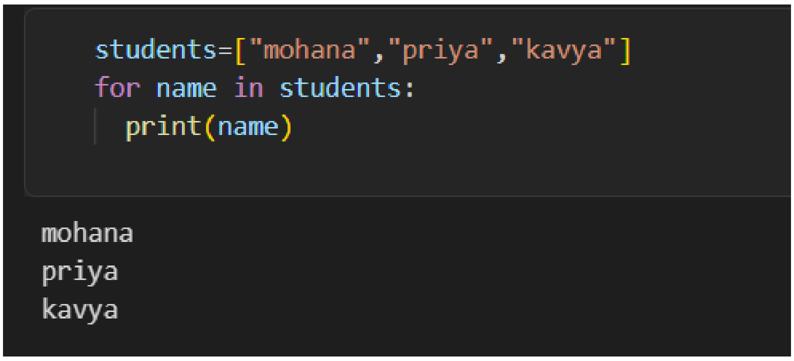
Q4. Display all student names using a loop

Aim: Display all student names using a loop.

Program:

```
students=["mohana","priya","kavya"]
for name in students:
    print(name)
```

Screenshot output:



```
students=["mohana","priya","kavya"]
for name in students:
  print(name)

mohana
priya
kavya
```

Logic:

1. Create a list containing student names.
2. Use a for loop to access each name in the list.
3. Store each name temporarily in the variable name.
4. Print the name using the print() function.
5. Repeat until all student names are displayed.

Q5. Create a function to add numbers

Aim: Create a function to add numbers.

Program:

```
def add_numbers(a,b):
    return a+b
num1 = int(input("Enter first number:"))
num2 = int(input("Enter second number:"))
result = add_numbers(num1 ,num2)
print("Sum=",result)
```

Screenshot output:

```
def add_numbers(a,b):
    return a+b
num1 = int(input("Enter first number:"))
num2 = int(input("Enter second number:"))
result = add_numbers(num1 ,num2)
print("Sum=",result)
```

```
Enter first number:4
Enter second number:6
Sum= 10
```

Logic:

1. Create a function named add_numbers() that accepts two numbers as parameters.
2. Add the two numbers inside the function.
3. Return the sum using the return statement.
4. Read two numbers from the user.
5. Call the function and pass the input values as arguments.
6. Store the returned result in a variable.
7. Display the sum.

Q6. Create a function to display student details

Aim: Create a function to display student details.

Program:

```
def student_details(name ,age ,branch):
    print("student Name:",name)
    print(" Age:",age)
    print("Branch",branch)
student_details("priya", 18, "CSM")
```

Screenshot output:

```
def student_details(name ,age ,branch):
    print("student Name:",name)
    print(" Age:",age)
    print("Branch",branch)
student_details("priya", 18, "CSM")
```

```
student Name: priya
 Age: 18
Branch CSM
```

Logic:

1. Create a function named display_student().
2. Pass student name, age and branch as parameters.
3. Display the student details using print().
4. Call the function with the required student information.
5. The function prints all student details on the screen.

Conclusion:

The task helped me understand python functions and lists I learned how to create functions,manage data using lists, and develop simple programs.